

Who's Involved in Crimes In Dallas

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2026-04-08

The main question I want to answer is who is most involved in crime incidents. This includes analyzing victims such as the type, race, gender, and ethnicity to understand patterns in crime involvement.

```
# 1. LOAD LIBRARIES
library(ggplot2)
```

Loading Data

```
setwd("~/rmarkdown data")
data <- read.csv("who.csv")

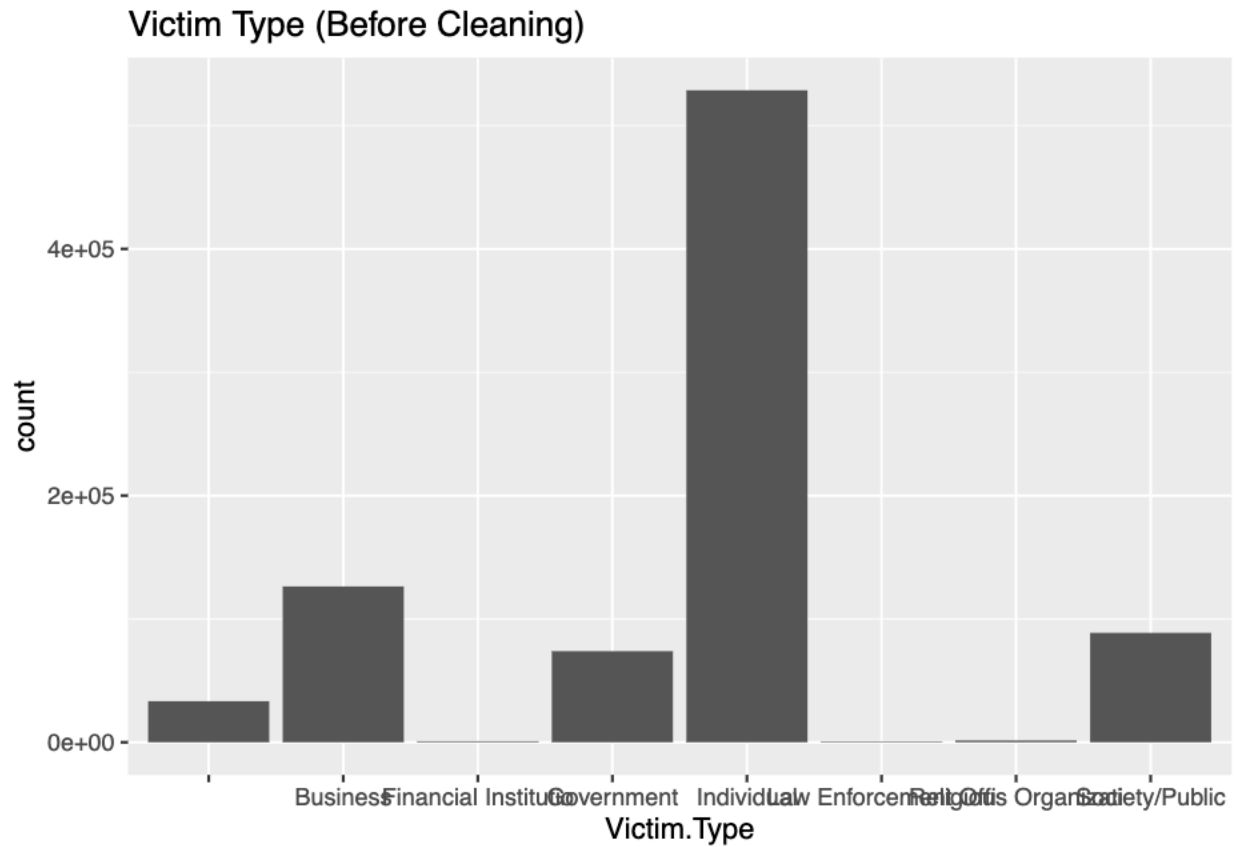
head(data)
```

```
##   Year.of.Incident      Victim.Race  Victim.Type  Victim.Ethnicity
## 1          2020 Hispanic or Latino  Individual  Hispanic or Latino
## 2          2020             White  Individual Non-Hispanic or Latino
## 3          2020                               Business
## 4          2020 Hispanic or Latino  Individual  Hispanic or Latino
## 5          2020 Hispanic or Latino  Individual  Hispanic or Latino
## 6          2020                               Society/Public
##   Victim.Gender
## 1           Male
## 2          Female
## 3
## 4          Female
## 5           Male
## 6
```

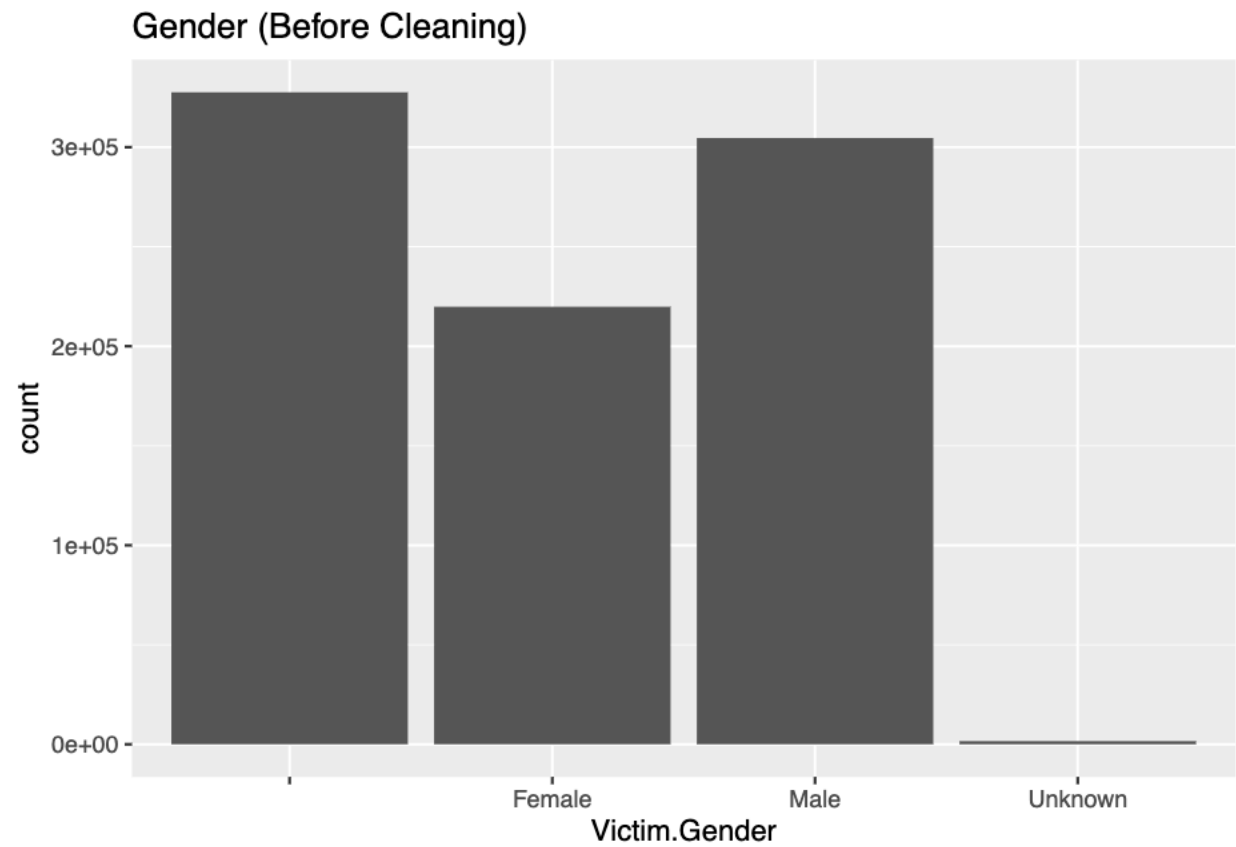
Before Cleaning

The dataset had several issues that needed to be fixed. Its main issue was that it was missing or had unclear values in the victim type, gender, race, and ethnicity columns. Many entries were blank spaces such as "" which don't mean anything when analyzing.

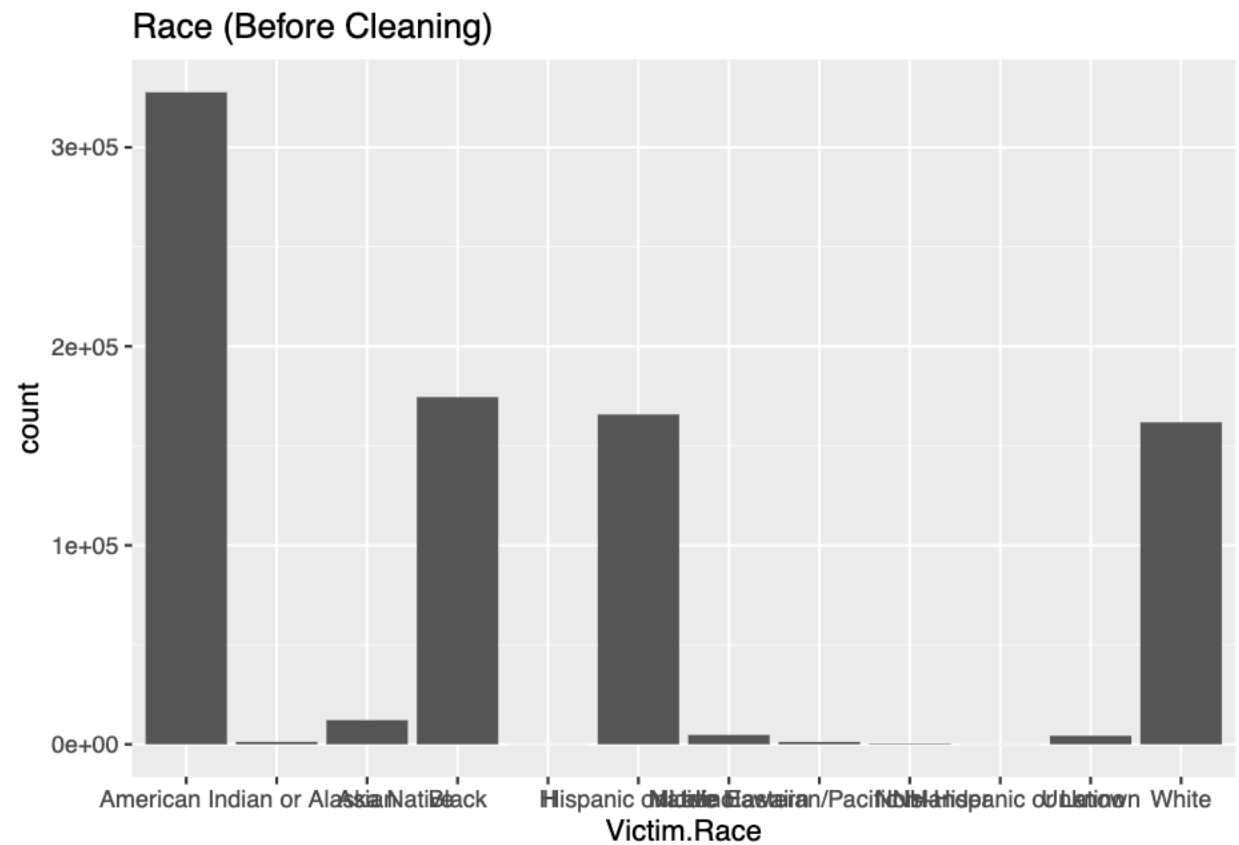
```
#victim type before fixing anything
ggplot(data, aes(x = Victim.Type)) +
  geom_bar() +
  labs(title = "Victim Type (Before Cleaning)")
```



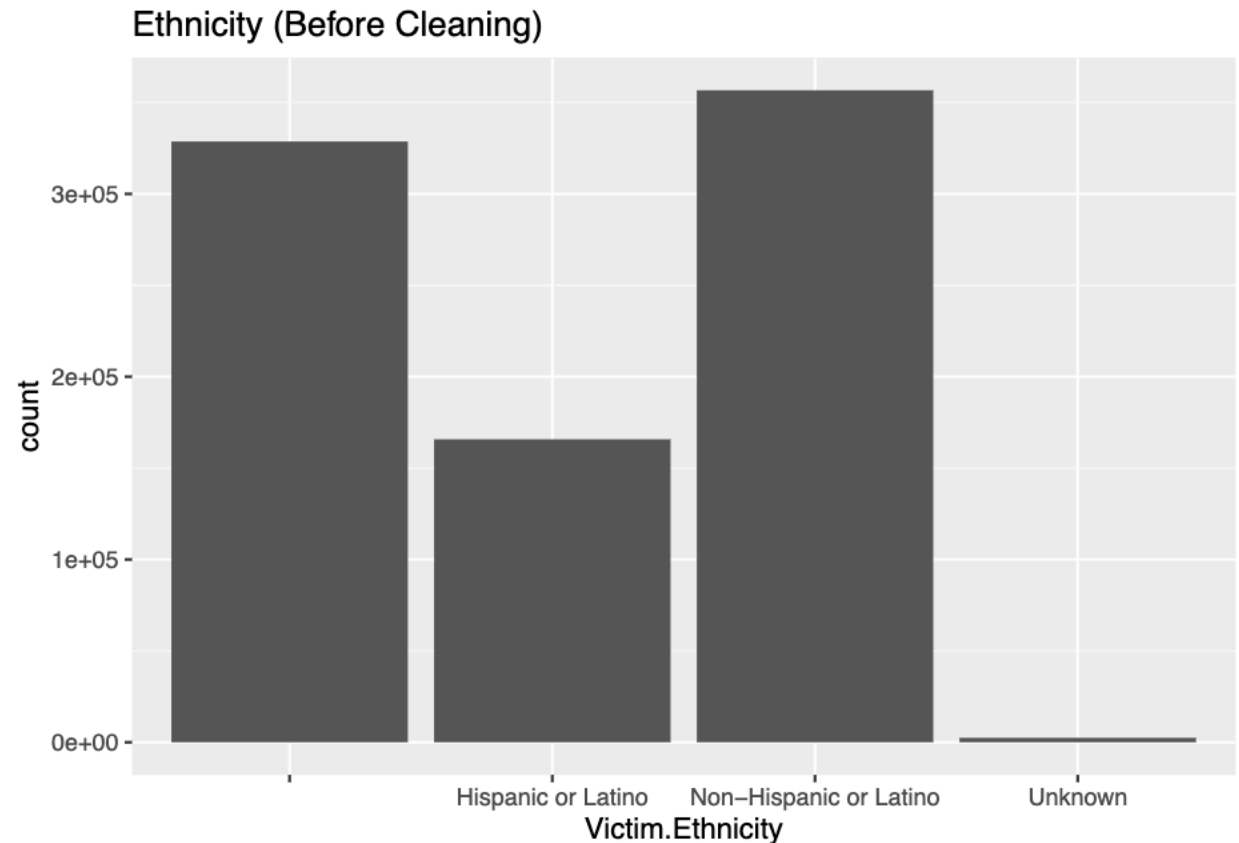
```
#victim gender  
ggplot(data, aes(x = Victim.Gender)) +  
  geom_bar() +  
  labs(title = "Gender (Before Cleaning)")
```



```
#victim race (have some blanks)  
ggplot(data, aes(x = Victim.Race)) +  
  geom_bar() +  
  labs(title = "Race (Before Cleaning)")
```



```
#Victim Ethnicity (lots of blanks too)
ggplot(data, aes(x = Victim.Ethnicity)) +
  geom_bar() +
  labs(title = "Ethnicity (Before Cleaning)")
```



Data Cleaning

To fix the missing or " " sections, I removed the blank values from the dataset. This step was necessary because keeping these entries would make the graphs confusing and harder to understand. By removing unclear or incomplete data, the dataset became more consistent.

```
clean_data <- data

## remove blanks like ("") and replacing them with NA.

#type
clean_data$Victim.Type[clean_data$Victim.Type == "" | clean_data$Victim.Type == " "] <- NA

#gender
clean_data$Victim.Gender[clean_data$Victim.Gender == "" | clean_data$Victim.Gender == " "] <- NA

#race
clean_data$Victim.Race[clean_data$Victim.Race == "" | clean_data$Victim.Race == " "] <- NA
clean_data$Victim.Race[clean_data$Victim.Race == "H"] <- "Hispanic or Latino"
clean_data$Victim.Race[clean_data$Victim.Race == "NH"] <- "Non-Hispanic or Latino"

#ethnicity
clean_data$Victim.Ethnicity[clean_data$Victim.Ethnicity == "" | clean_data$Victim.Ethnicity == " "] <- NA
```

After Cleaning

After cleaning, the graphs of each variable were easier to understand. The results presented more of an accurate pattern in the data since only valid entries were included. I've also changed the angle of the text for the y-axis since some of the text were way too long. Overall, the cleaning process has improved the quality of the dataset and made the analysis more reliable.

```
# check summary stats to see if missing things were removed
summary(clean_data)
```

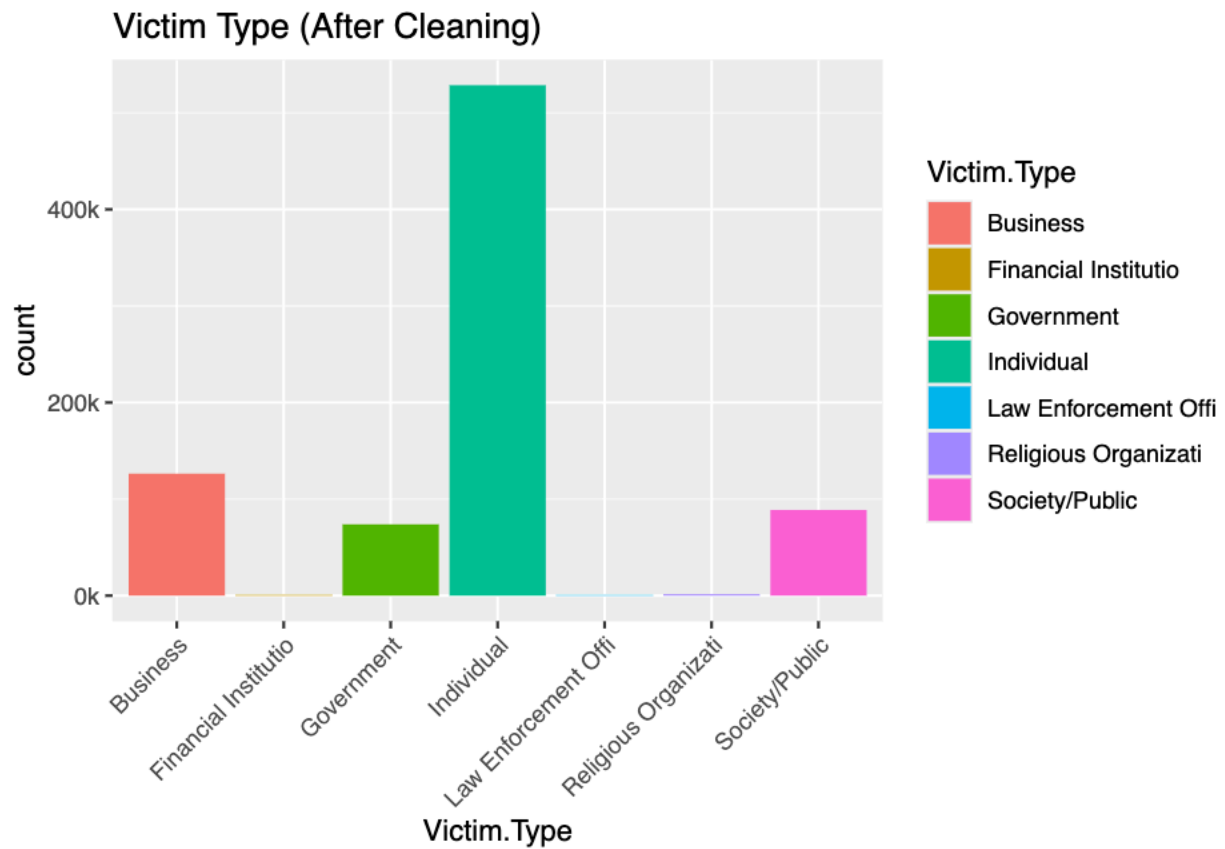
```
## Year.of.Incident Victim.Race      Victim.Type      Victim.Ethnicity
## Min.      :2020      Length:853536      Length:853536      Length:853536
## 1st Qu.:2021      Class :character      Class :character      Class :character
## Median :2023      Mode  :character      Mode  :character      Mode  :character
## Mean      :2023
## 3rd Qu.:2024
## Max.      :2026
## Victim.Gender
## Length:853536
## Class :character
## Mode  :character
##
##
##
```

```
colSums(is.na(clean_data))
```

```
## Year.of.Incident      Victim.Race      Victim.Type      Victim.Ethnicity
##              0              327657              33282              328678
##      Victim.Gender
##              327560
```

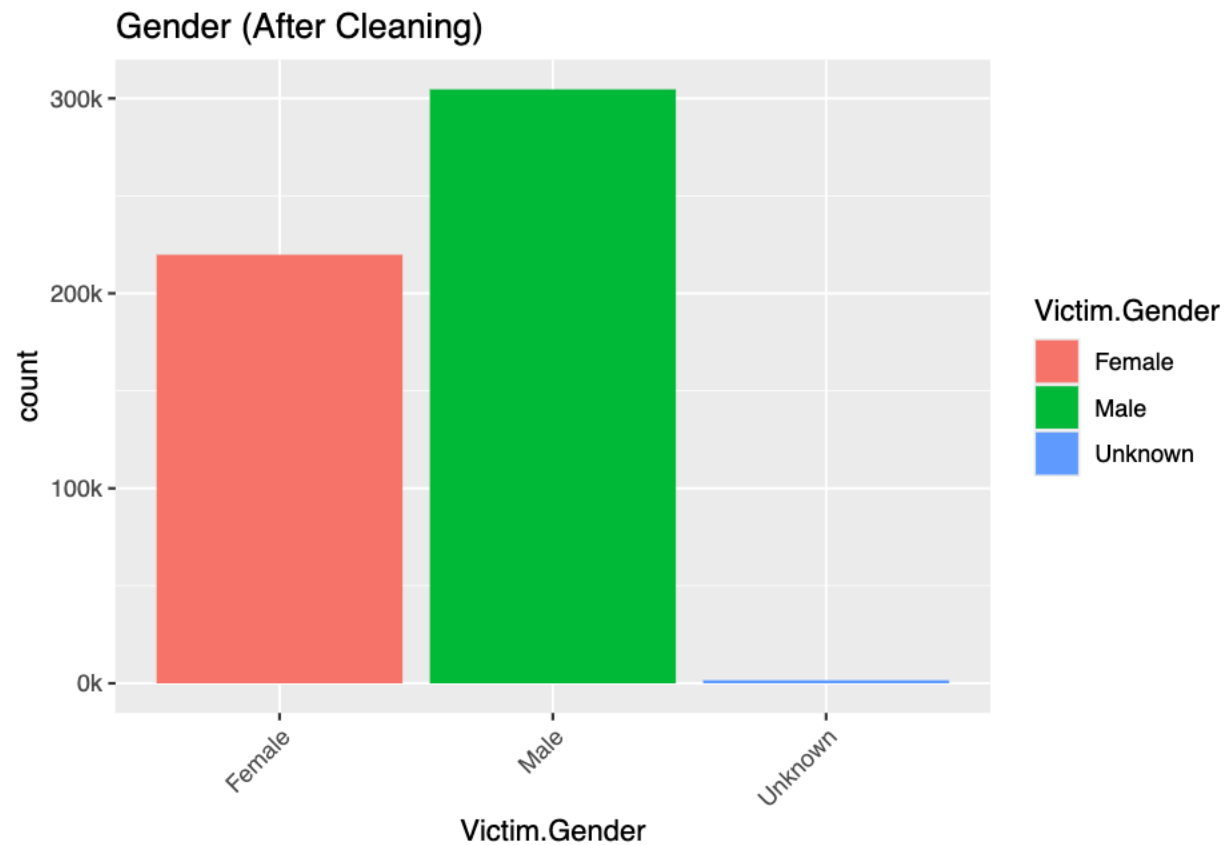
```
# type
type_data <- clean_data[!is.na(clean_data$Victim.Type), ]

ggplot(type_data, aes(x = Victim.Type, fill = Victim.Type)) +
  geom_bar() +
  labs(title = "Victim Type (After Cleaning)") + theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



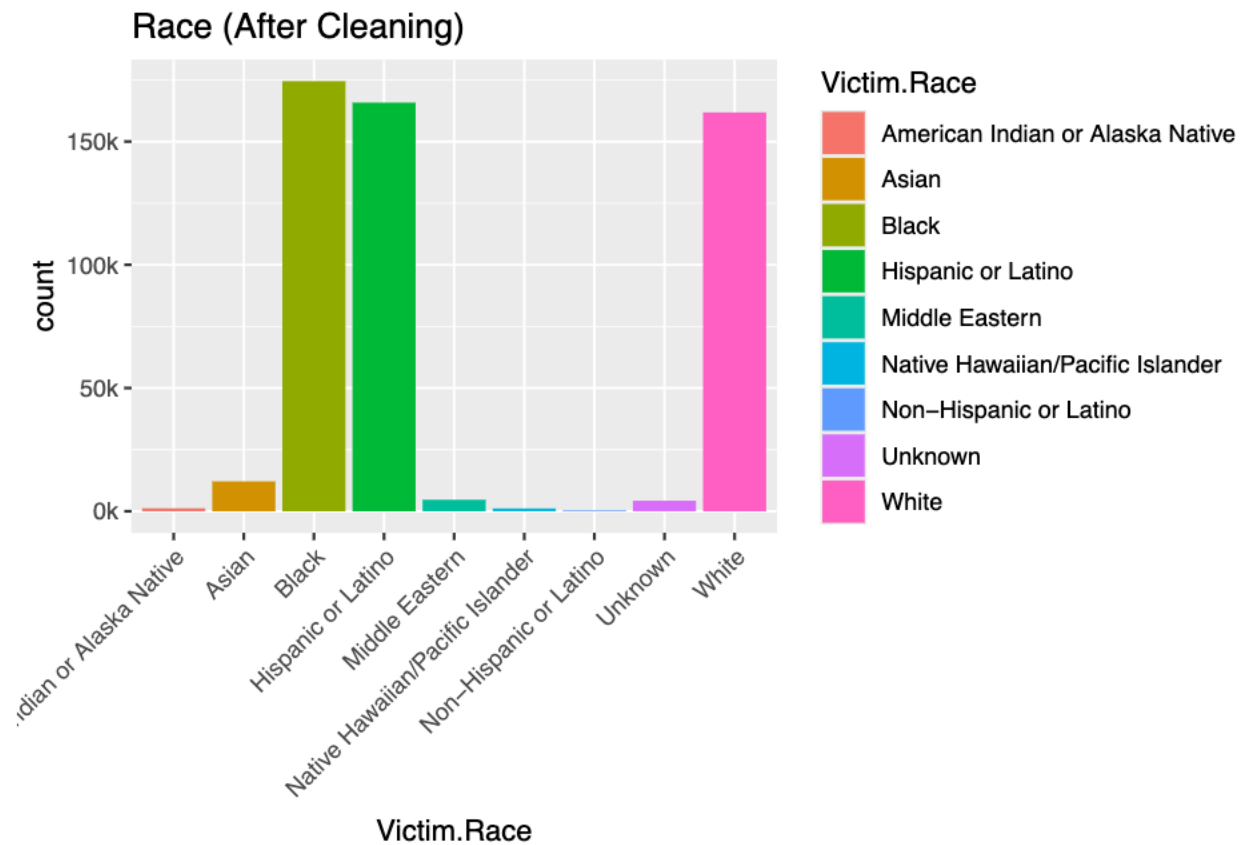
```
# gender
gender_data <- clean_data[!is.na(clean_data$Victim.Gender), ]

ggplot(gender_data, aes(x = Victim.Gender, fill = Victim.Gender)) +
  geom_bar() +
  labs(title = "Gender (After Cleaning)") + theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
```



```
# race
race_data <- clean_data[!is.na(clean_data$Victim.Race), ]

ggplot(race_data, aes(x = Victim.Race, fill = Victim.Race)) +
  geom_bar() +
  labs(title = "Race (After Cleaning)") + theme(axis.text.x = element_text(angle = 45, hjust = 1)) + sc
```



```
# ethnicity
ethnicity_data <- clean_data[!is.na(clean_data$Victim.Ethnicity), ]

ggplot(ethnicity_data, aes(x = Victim.Ethnicity, fill = Victim.Ethnicity)) +
  geom_bar() +
  labs(title = "Ethnicity (After Cleaning)") + theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

